

Smart Monitoring System for Hospital Wards

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Context and Physical Space

- **Objective:** Improve patient safety through real-time IoT-enabled monitoring.
- **Deployment:** Standard hospital rooms equipped with smart beds and sensors.
- **Data Hub:** Central nurse's station monitors vital signs on a dashboard.
- **Infrastructure:** Wireless data transmission with redundancy for uninterrupted service.
- **Key Features:** Backup servers ensure data security during technical failures.

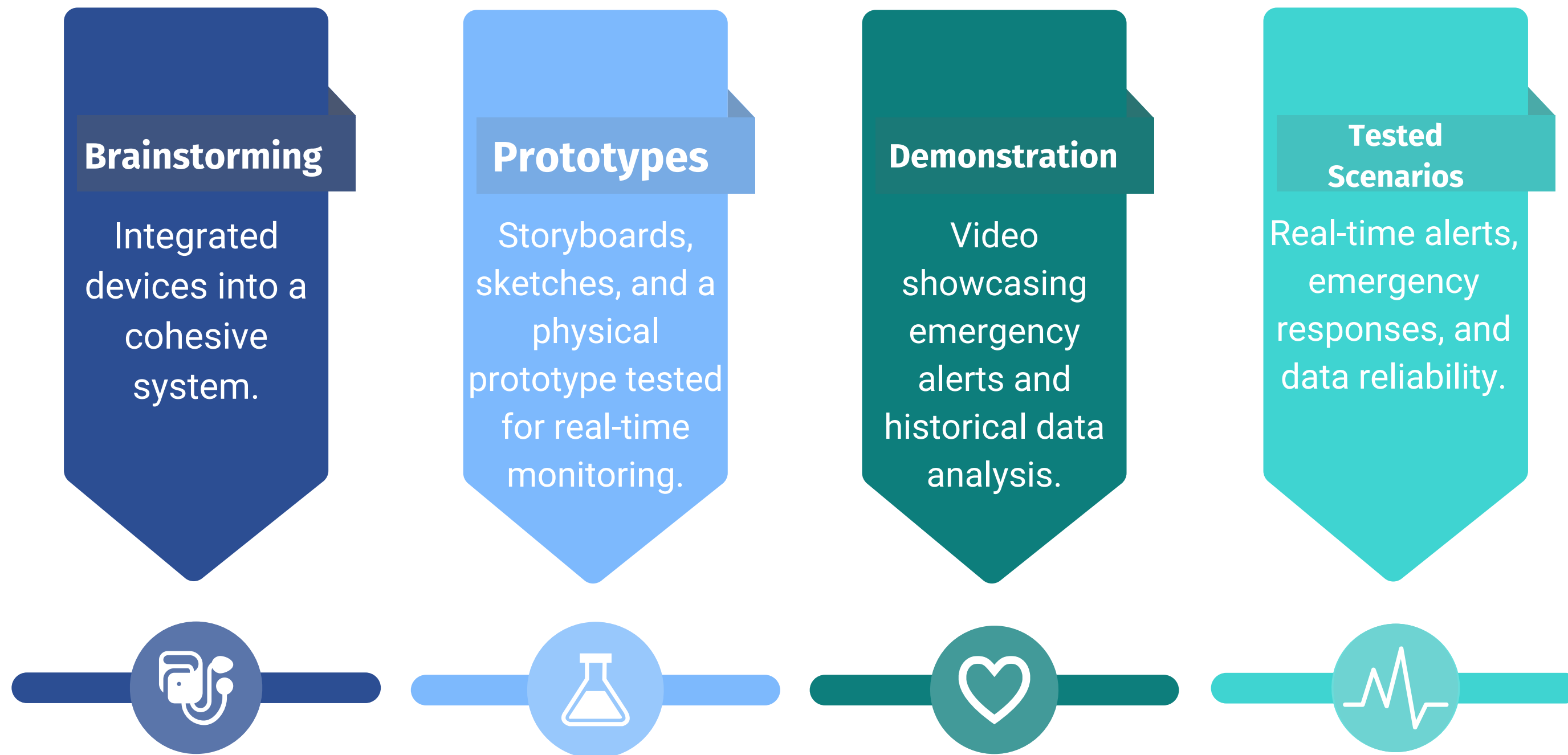


Initial Ideas

- **Smart Bed:** Continuous vitals monitoring and abnormality alerts via IoT sensors.
- **Smart Clipboard:** Digitized patient records with AI-based symptom analysis.
- **Smart Cap:** Non-invasive EEG monitoring for brain activity.
- **Room Management:** AI and cameras monitor room occupancy and prevent overcrowding.



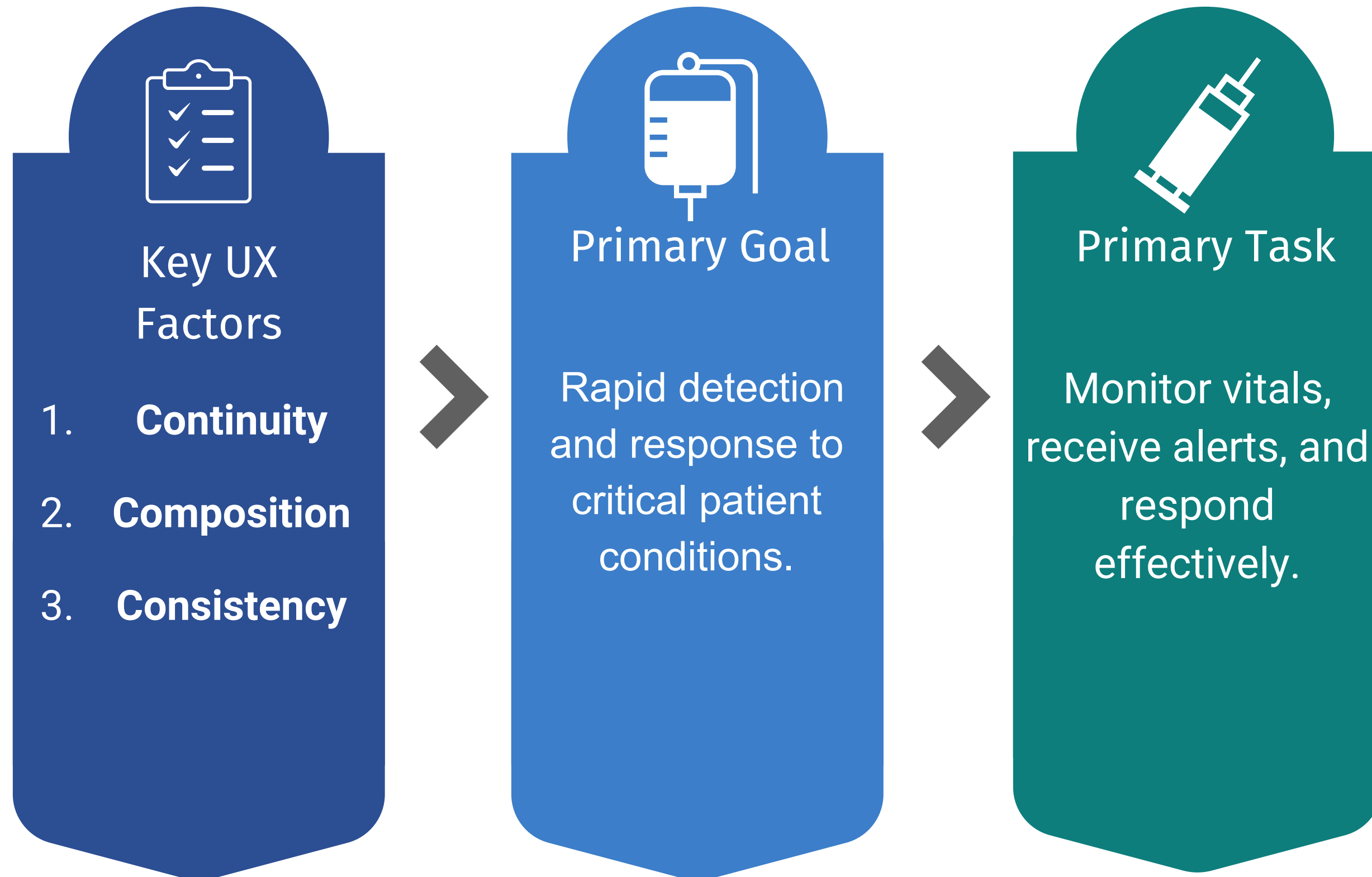
Prototype Development



Prototype Features

Physical Object	Activity	Smart Transformation	Interaction Experience
Smart Bed	Monitor Vitals	IoT sensors, alerts	High visibility & alerts
Smart Cap	EEG Monitoring	EEG sensors	Moderate interaction
Smart Clipboard	Data logging, analysis	Cloud AI feedback	Medium complexity
Patient Management	Crowd Monitoring	AI and cameras	Moderate visibility

User Experience (UX)

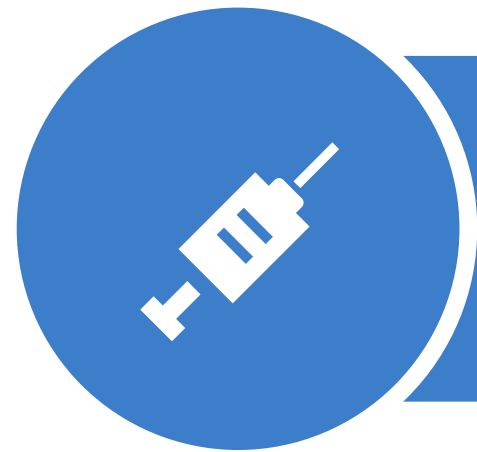


UX Metrics



Continuity

Data flows are automated but require accuracy checks.



Composition

Tools are user-friendly with slight inconsistencies in alert displays.



Consistency

Consistent design with minor adaptations needed for module variations.



Usability Pilot Testing

Overall Setting

The usability pilot will be conducted in a hospital simulation lab designed to replicate real hospital workflows.

Goal

The usability pilot will be conducted in a hospital simulation lab designed to replicate real hospital workflows. The test setting will include complete ward setups with functional smart beds and devices.



Usability Pilot Testing

Participants

12 participants including:

- 6 **nurses** (primary users for real-time monitoring)
- 3 **physicians** (secondary users for data analysis)
- 2 **IT staff** (for technical setup and maintenance)
- 1 **hospital administrator** (for managerial insights and reporting)

Tasks

- System login and dashboard configuration
- Real-time monitoring of vital signs.
- Responding to simulated critical alerts.
- Using predictive analytics for risk assessment.
- Retrieving historical patient data and making informed decisions.



Deliverables

01

PowerPoint Presentation: A summarized presentation showcasing the project phases, design choices, and key results from usability testing.

02

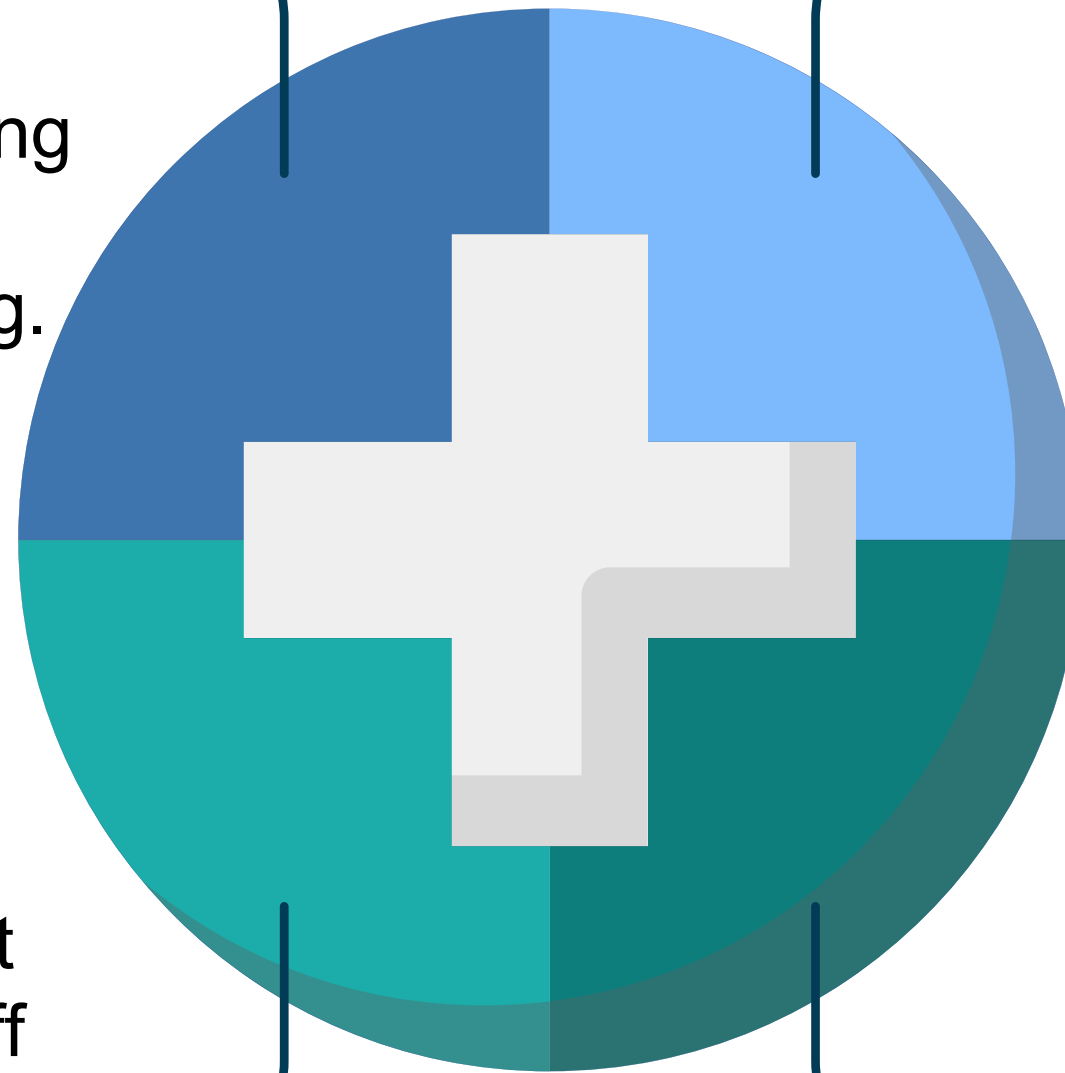
Physical Prototype: A working prototype of the smart monitoring system was developed and tested for initial feedback

03

Video Prototype: : A storyboard-based video demonstrating the system's workflow, including patient monitoring, alert reception, and staff intervention steps.

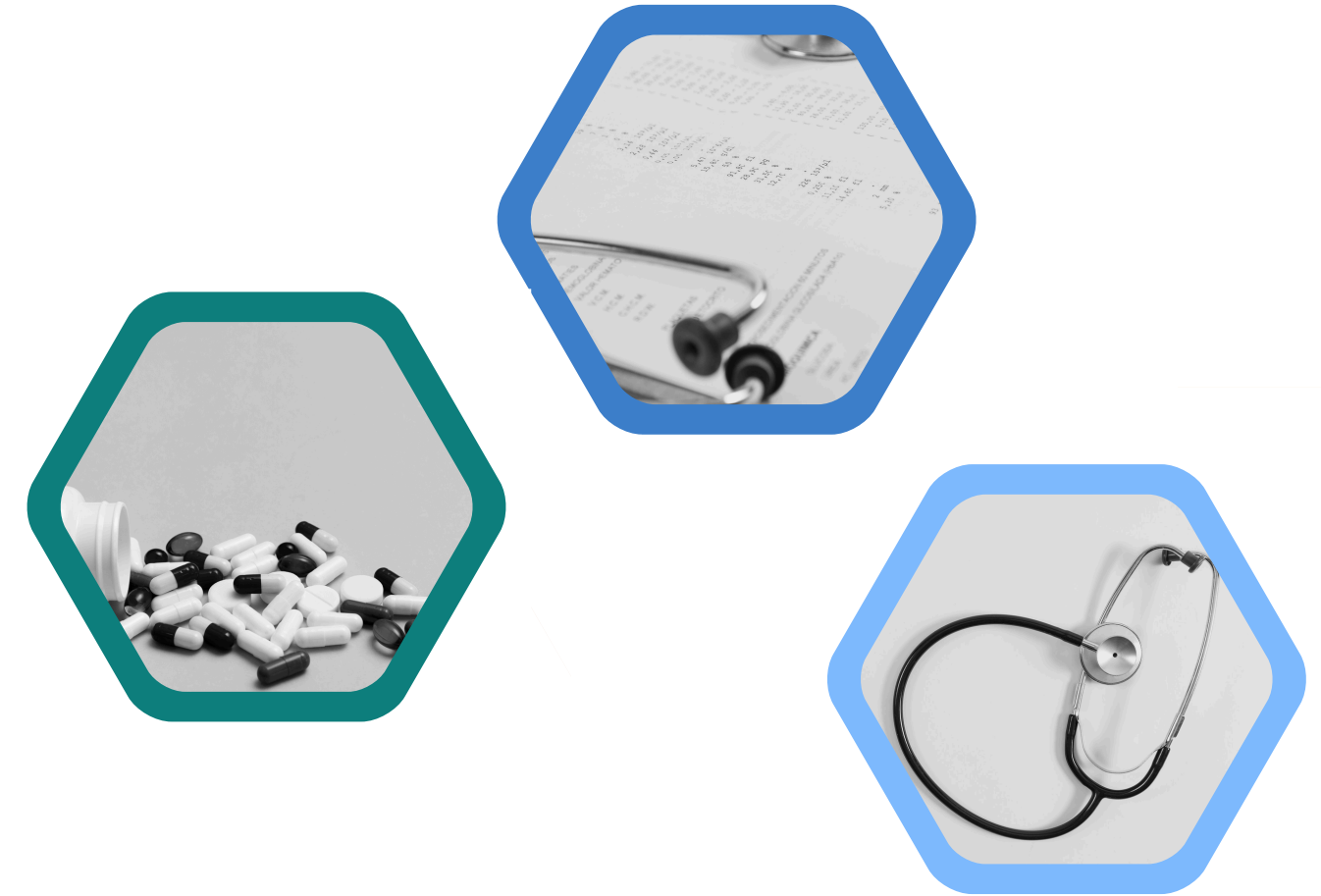
04

Final Report Document: This comprehensive report detailing the system design, testing methodology, and final recommendations



Conclusion

The Smart Monitoring System for hospital wards improves patient care by offering real-time monitoring, predictive analytics, and proactive alerts. It enhances patient safety, reduces response times, and provides clear data to medical staff.



Pilot tests have shown its effectiveness, and with proper training and optimization, it has the potential to become a standard in modern healthcare.

